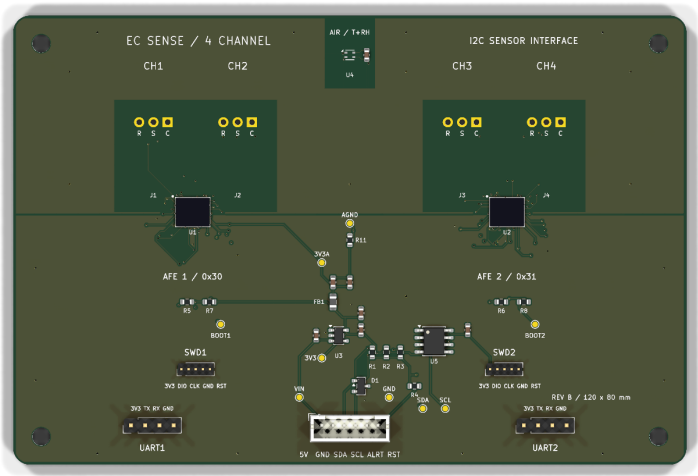


EC Sense - four-channel I2C PCB

Revision B | 120 x 80 mm | Fabrication and assembly | 16 September 2026

Fabrication specification

Board	120.00 x 80.00 mm, R3 corners
Construction	4 copper layers, 1.6 mm FR-4, ENIG
Track / space	0.075 / 0.075 mm minimum
Fine vias	0.20 mm finished drill / 0.40 mm pad
Stitching vias	0.30 mm finished drill / 0.60 mm pad
Mounting	4 x 3.2 mm NPTH; 110 x 70 mm pattern
Assembly	Components on both sides; tented vias



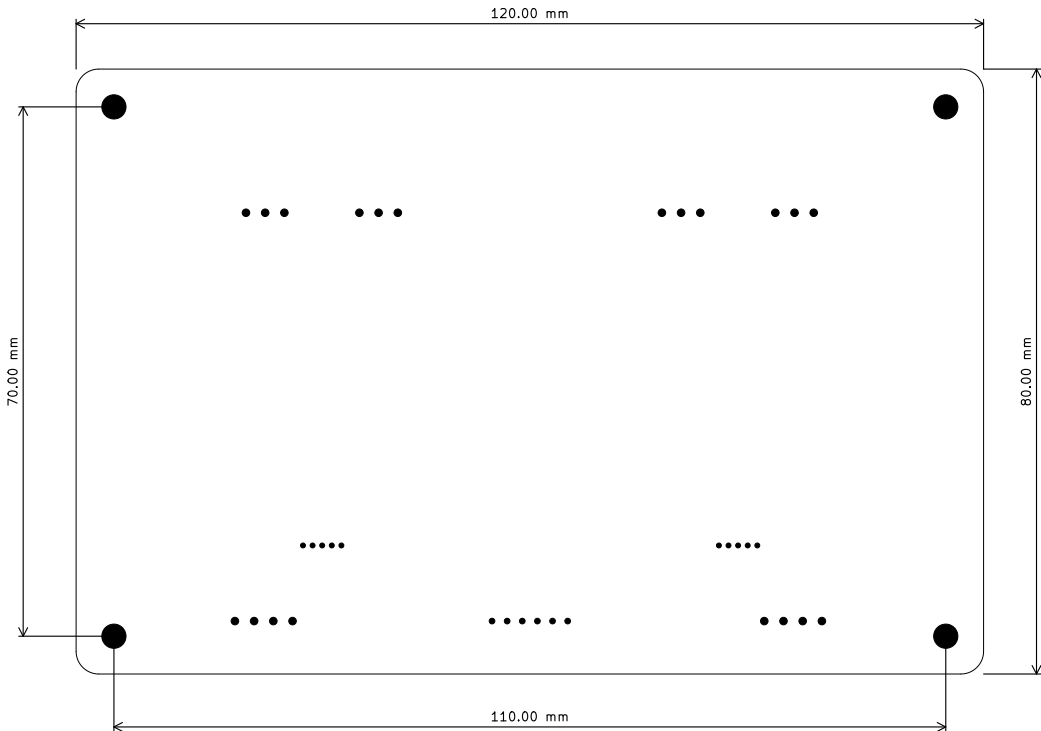
Illustrative 3D view; land patterns govern assembly.

Assembly: Fit all capacitors and R11. Leave JP1-JP4 open. Each sensor socket uses three Mill-Max 0548-0-15-15-21-27-10-0 receptacles. Insert gas sensors after soldering and cleaning.

Checks: Zero DRC violations, unconnected items, schematic mismatches and ERC violations. This is an untested hardware prototype: firmware, gas-specific setup, zero/span calibration and bench validation remain required.

Outputs: Use the Gerbers and separate PTH/NPTH drills for fabrication. Absolute KiCad origin is shared across outputs. Bottom assembly is viewed from below; do not mirror Gerbers. See PCB_README.md for power, connectors and bring-up.

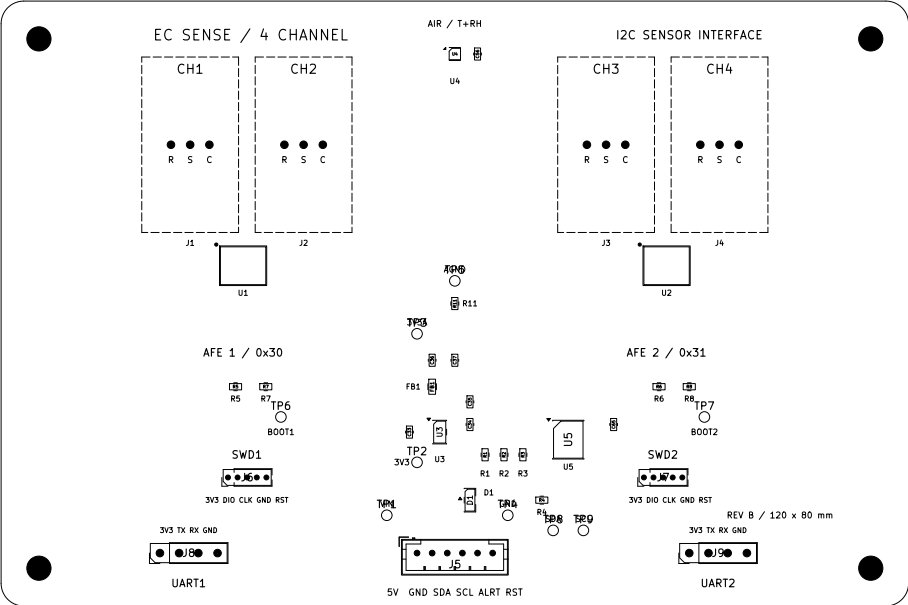
Mechanical - dimensions in millimeters



4 LAYERS / 1.6 mm FR-4 / ENIG
MIN TRACK / SPACE 0.075 mm; VIA DRILL 0.20 mm
4 x 3.2 mm NPTH / R3 CORNERS

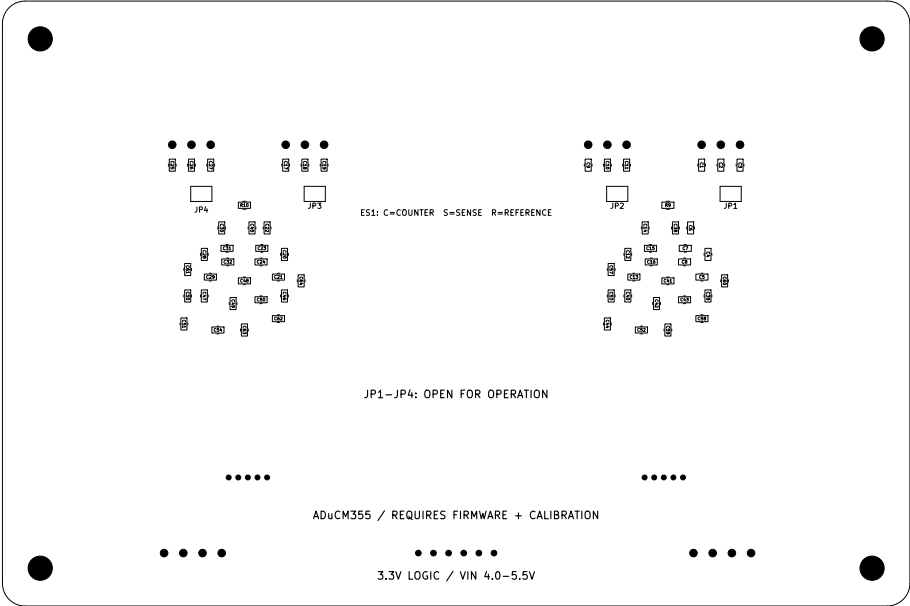
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File: ECSense_4CH_I2C.kicad_pcb		
Title: EC Sense 4-channel I2C sensor interface		
Size: A4	Date: 2026-09-16	Rev: B
KiCad E.D.A. 10.0.5	Id: 1/1	

Assembly - front



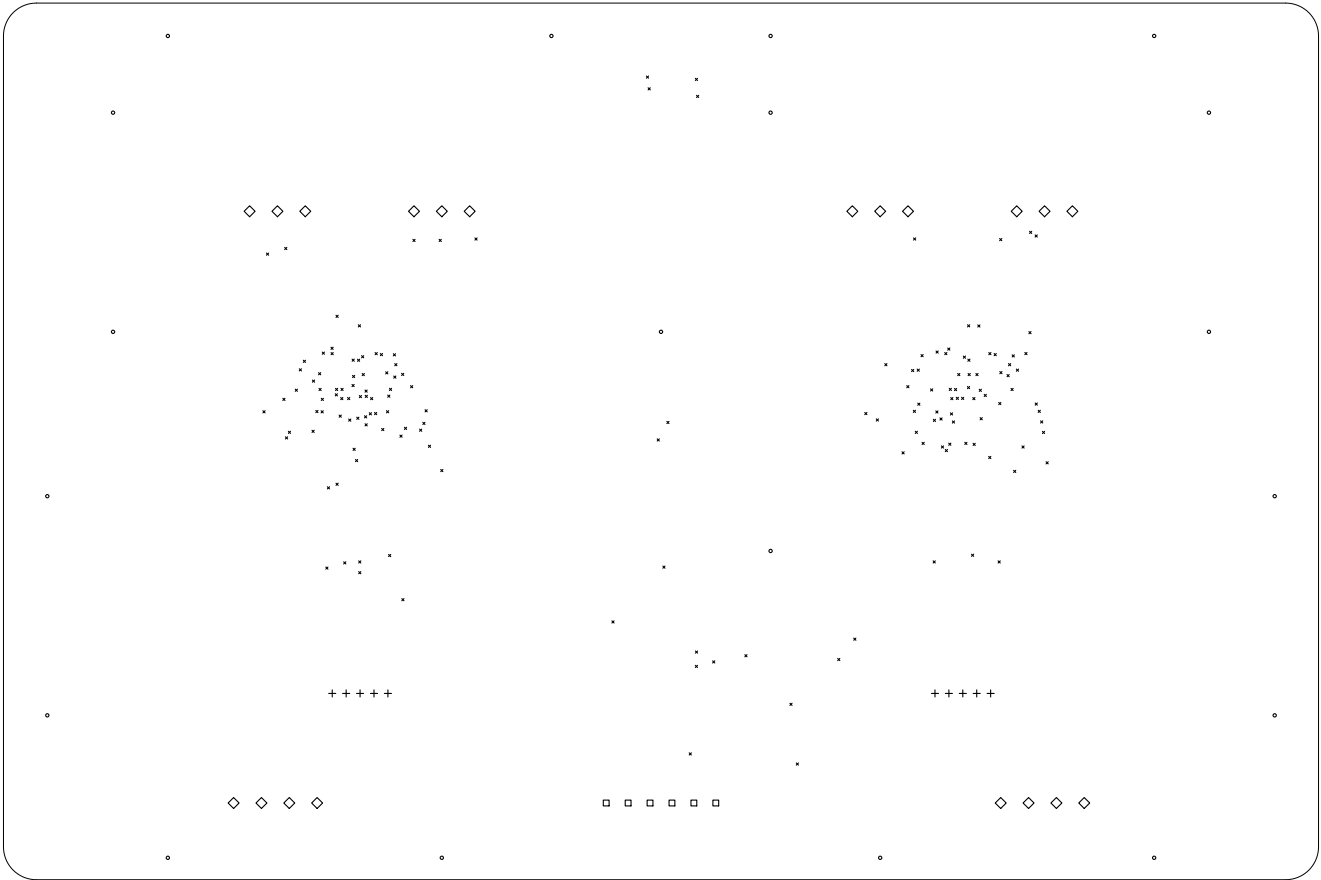
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Title: EC Sense 4-channel I2C sensor interface		
Size: A4	Date: 2026-09-16	Rev: B
KiCad E.D.A. 10.0.5	Id: 1/1	

Assembly - bottom, viewed from below



Sheet:		
File: ECSense_4CH_I2C.kicad_pcb		
Title: EC Sense 4-channel I2C sensor interface		
Size: A4	Date: 2026-09-16	Rev: B
KiCad E.D.A. 10.0.5	Id: 1/1	

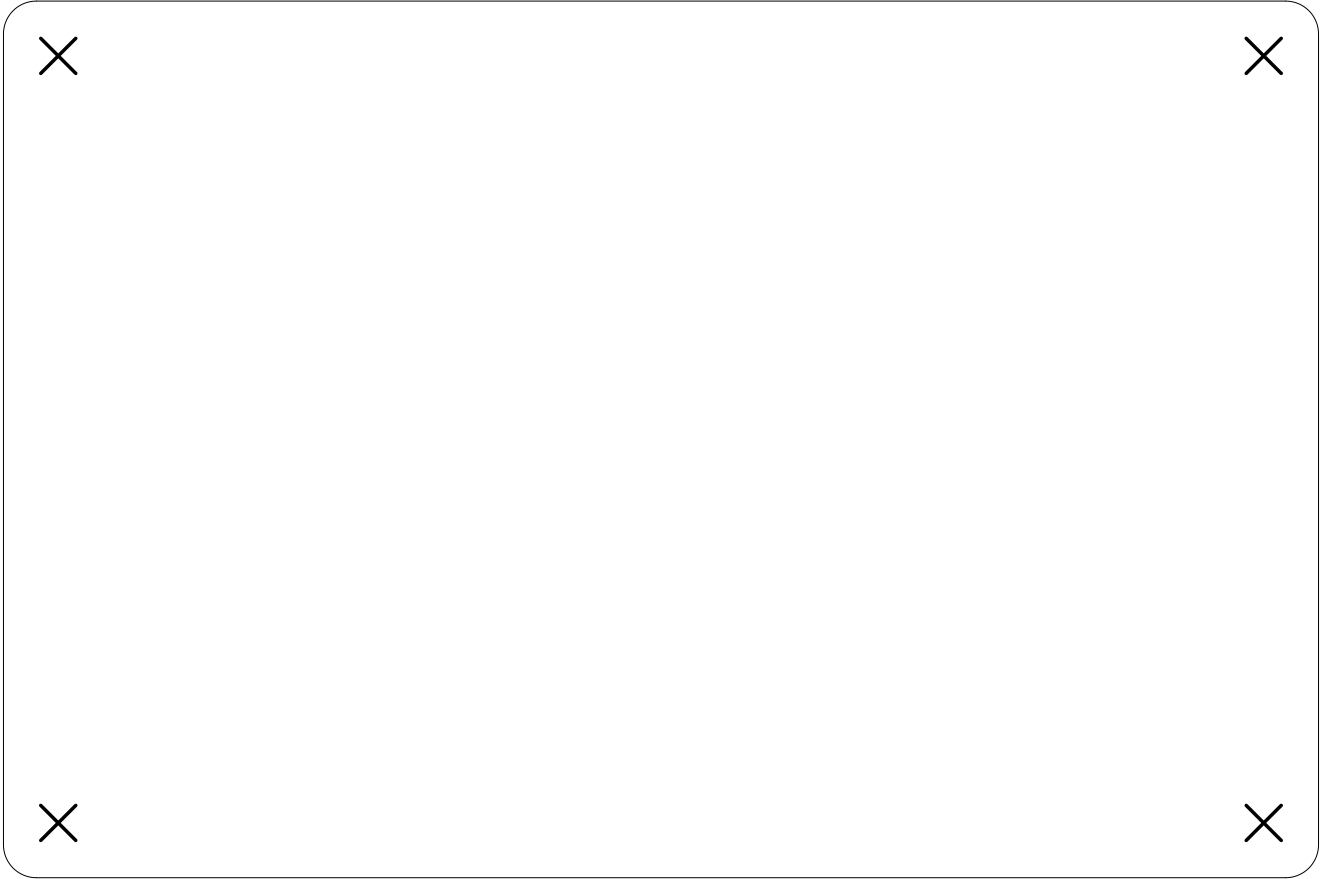
Plated drill map



Drill Map:

- 0.200mm / 0.0079" (161 holes)
- 0.300mm / 0.0118" (19 holes)
- + 0.650mm / 0.0256" (10 holes)
- ◻ 0.750mm / 0.0295" (6 holes)
- ◊ 1.000mm / 0.0394" (20 holes)

Nonplated drill map



Drill Map:

× 3.200mm / 0.1260" (4 holes) (not plated)